

Methodological Review and Redesign of My First Research Project

Dye removal from model textile wastewater by coagulation and sorption (2019)

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Work under review	«Purification of Wastewater from Dye Production» («Очистка сточных вод красильного производства»), chemistry, grade 10
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Abstract

Background. In 2019, as a grade 10 student, I carried out an experimental study on the removal of textile dyes from synthetic wastewater by coagulation with iron(III) and aluminium salts, by sorption onto sulfonated coal, and by the two processes applied in sequence. The work was supervised jointly by a school chemistry teacher and a university faculty member, and received a Diploma of the Second Degree at an international student research competition.

Objective. To review the design, measurement, analysis, and reporting of that study against the standards applied in the peer-reviewed water treatment literature, and to specify a redesign that would bring the same research question to publishable standard.

Materials and methods. The review works from the archived manuscript and its tables and figures. Thirteen appraisal criteria were applied, covering design architecture, analytical definition, dosing specification, procedural specification, replication, control of experimental variables, adsorption characterisation, sorbent characterisation, mechanistic attribution, endpoint coverage, matrix representativeness, techno-economic basis, and reporting practice. Each criterion records what the paper reports and what a complete report of that design would additionally contain.

Findings. The study design supports comparative statements about the ordering of treatment performance, and this ordering was consistent across four dye application classes. Quantitative statements about the magnitude of removal require a calibrated measurement chain, a stated uncertainty, and replication, which a design of this construction does not

provide. Each of the thirteen criteria identifies reporting elements that a complete account of this design would contain and that the manuscript does not include, the largest being analytical method definition, dose expressed in concentration units, replication, and adsorption capacity.

Conclusions. The redesign specified here retains the original research question and the two-stage treatment architecture, and changes what is measured and how it is reported. It is set out as a seven-phase work programme requiring roughly ten to twelve weeks of part-time laboratory work, with three procedures requiring access to a university chemistry department.

Keywords: methodological review, research design, textile wastewater, reactive dyes, coagulation, adsorption, analytical validation, reporting standards, student research.

1. Introduction

1.1 Rationale

I completed this project in grade 10 and did not return to it between 2019 and the writing of this review. The reasoning behind its design choices was not recorded at the time. This review reconstructs that reasoning and examines the study against the standards I have since learned to apply.

The reasoning behind a design choice is invisible in a published result. A reader of the 2019 paper sees that four model waters were prepared and that optical density was measured; the reader does not see why those choices were made, what they permit the study to conclude, and what a different choice would have permitted instead. This review makes that reasoning explicit at each point.

The intended readership is students planning a first experimental project, and researchers assessing how I work.

1.2 Objectives

The review has three objectives. It documents the design and execution of the 2019 study in a form that can be assessed by a reader without access to the original Russian manuscript. It assesses the reporting completeness of that study against criteria drawn from the water treatment and analytical chemistry literature. It specifies a redesign of the same investigation, with a work programme and resource requirement.

2. Materials and scope

2.1 The work under review

The stated aim of the 2019 study was to develop an effective method for treating dye-bearing wastewater using sorbents with sufficient sorption capacity, availability, and low cost. The manuscript set four tasks: to examine the use of coagulants against reactive dyes, to examine sulfonated coal as a low-cost sorbent, to determine the parameters governing the sorption process, and to produce recommendations for the application of such materials.

Four synthetic model waters were prepared. Each combined two or three commercial dyes drawn from a single application class: direct (model water 1), acid (model water 2), disperse (model water 3), and reactive (model water 4). These classes differ in charge, water solubility, and mechanism of attachment to the fibre. Direct and acid dyes are anionic and water-soluble. Reactive dyes are anionic and form covalent bonds with cellulose during dyeing, with fixation efficiency for cotton typically between 50 and 80 per cent, so that between a fifth and a half of the applied dye is discharged as unfixed or hydrolysed material [8]. Disperse dyes are effectively non-ionic and sparingly soluble, and are carried as a fine dispersion.

Three treatments were applied to each model water. Coagulation used an iron(III) salt, and separately an aluminium salt, dosed at 1.75 mL into 300 mL of coloured water, with the floc filtered off before measurement. Sorption passed 100 mL portions through a packed bed of sulfonated coal supported on a glass filter, with the bed mass increased until the filtrate met the target. The third treatment applied coagulation and sorption in sequence.

Colour was followed photometrically and reported as optical density, with transmittance and a percentage degree of dye removal derived from it. Solution pH was recorded before and after each treatment. Supporting characterisation of the sorbent gave moisture content 2.1 per cent, ash content 30 per cent, and intrinsic pH 5.41. A dose-response series recorded the optical response at eleven sorbent masses from 0 to 10 g.

Reported outcomes were mean dye removal of 66 to 67 per cent for coagulation alone, 80 per cent for sorption alone, and 98.60 per cent for the two applied in sequence. Solution pH moved from acidic toward near-neutral under every treatment. Twenty-five sources were cited, drawn from Russian-language monographs on adsorption, colloid chemistry, ion exchange, and water treatment engineering.

2.2 Materials available to this review

This review works from the archived manuscript, including its eight tables and one figure, and from the competition diploma. Laboratory notebooks, raw instrument records, calibration records, and reagent specifications from 2018 and 2019 were not available. Where a criterion in Section 4 concerns information that may have existed in the laboratory without entering the manuscript, this is stated in the relevant entry.

2.3 Scope limits

This review does not repeat the experiments, does not reanalyse raw measurements, and does not assess the 2019 study against the standards applied to school research projects, which it met. The comparator throughout is the reporting standard of the peer-reviewed water treatment literature.

3. Appraisal framework

3.1 Criteria

Thirteen criteria were applied. Criteria 1 to 4 concern the design and whether an independent worker could reproduce the experiment from the text. Criteria 5 to 9 concern whether the

measurements support the interpretations placed on them. Criteria 10 to 13 concern whether the study characterises the process it recommends and reports it adequately.

C1. Design architecture. Whether the arrangement of treatments and controls permits the contribution of each process stage to be separated.

C2. Analytical definition. Whether the measured quantity is defined: wavelength, instrument, path length, reference solution, calibration curve, linear range, and limits of detection and quantification.

C3. Dosing and reagent specification. Whether reagent identity, grade, stock concentration, and applied dose in concentration units are stated for every substance added.

C4. Procedural specification. Whether mixing regime, contact time, settling time, filtration medium, and temperature are stated. For coagulation the reference procedure is ASTM D2035 [1], which specifies systematic evaluation of the variables encountered in coagulation and flocculation followed by gravity settling.

C5. Replication and uncertainty. Whether each condition was measured more than once, and whether a measure of spread is reported.

C6. Control of experimental variables. Whether variables identified as governing the process were varied under control, in particular pH for hydrolysing metal coagulants.

C7. Adsorption characterisation. Whether contact time series and equilibrium isotherms were determined, and whether capacity is reported in mass of adsorbate per mass of adsorbent. The reference models are the pseudo-first-order and pseudo-second-order kinetic forms [2], the Langmuir [3] and Freundlich [4] isotherms, and the intraparticle diffusion analysis of Weber and Morris [5].

C8. Sorbent characterisation. Whether the properties governing sorbent behaviour are reported: particle size, specific surface area, point of zero charge, surface functional groups, and the provenance of each figure.

C9. Mechanistic attribution. Whether the mechanism invoked is supported by measurement, and whether alternative mechanisms were tested against.

C10. Endpoint coverage and mass balance. Whether the endpoint set covers the removal target, the residual load, the treatment reagents, and the secondary waste.

C11. Matrix representativeness. Whether the tested matrix corresponds to the effluent the process is recommended for, and whether the difference is stated.

C12. Techno-economic basis. Whether cost claims in the objectives are supported by a calculation from measured doses and material prices, and whether reuse of the material is established.

C13. Reporting and source practice. Whether every cited source is used at a specific point for a specific claim, whether the source base includes current peer-reviewed literature, and whether raw records are retained and made available.

3.2 Procedure

Each criterion was applied to the manuscript in turn. For each, the review records the elements the manuscript reports, and the elements a complete report of the same design would additionally contain. Section 4 presents these entries and Section 4.14 summarises them in tabular form. Section 5 states the classes of conclusion the 2019 design supports. Section 6 converts each gap into a specified procedure.

4. Findings

4.1 C1: Design architecture

The manuscript reports coagulation alone, sorption alone, and the two in sequence, applied to the same four model waters, with untreated water measured for every condition and treated samples referred to clean water.

This arrangement separates the contribution of each stage and then measures the sequence, which corresponds to the configuration used in industrial effluent treatment, where a physicochemical stage removes bulk load and a polishing stage handles the residue. Measuring each stage alone gives the basis for attributing the combined result between them.

The comparison across four application classes gives the design the capacity to distinguish behaviour general to dyes from behaviour specific to a charge class. Under a charge-dependent mechanism, the non-ionic disperse set would be expected to respond differently from the three anionic sets.

The dose-response series in the manuscript records eleven sorbent masses across a continuous range, and the coagulant series records five volumes. Both identify where the response is steep and where it flattens.

A complete report of this design would additionally state which model water the sorbent dose-response series describes, and would analyse the four application classes against each other rather than reporting a mean across them.

4.2 C2: Analytical definition

The manuscript reports optical density for every condition, derived transmittance, and a derived percentage degree of dye removal, with clean water named as the comparison.

A complete report would state the wavelength at which absorbance was measured, the instrument and its path length, and the solution used to zero it. It would present a calibration curve relating absorbance to known dye concentration, with the regression equation, the coefficient of determination, the concentration range over which the response is linear, and the limits of detection and quantification. It would express results as concentration in mg/L, with absorbance as the intermediate quantity.

These elements determine what a percentage removal figure means. Each model water contains two or three dyes with separate absorption maxima, so a single-wavelength reading integrates overlapping spectra and any turbidity present, and cannot be attributed to an

individual component. A calibration curve in the record lets a reader reconstruct the conversion from absorbance to concentration. The instrument settings and any calibration performed in 2019 may have existed in the laboratory without entering the manuscript.

4.3 C3: Dosing and reagent specification

The manuscript specifies 1.75 mL of coagulant added to 300 mL of coloured water, identifies the coagulants as an iron(III) salt and an aluminium salt, and specifies sorbent masses in grams against a stated sample volume of 100 mL.

A complete report would name the specific compound, the supplier, and the reagent grade for each coagulant, state the concentration of each stock solution, and express the applied dose as milligrams of iron or aluminium per litre of treated water. It would express sorbent dose in grams per litre and dye content in milligrams per litre.

Dose expressed in concentration units transfers between laboratories and between treated volumes. Expressed this way, the sorbent requirement recorded in the manuscript, 7.9 to 8.8 g per 100 mL, becomes 79 to 88 g/L, which enters a cost calculation directly and can be compared against dosing ranges used in practice.

4.4 C4: Procedural specification

The manuscript specifies the sample volumes, the sequence of operations, the filtration of floc before measurement, and the passage of the sample through a packed sorbent bed supported on a glass filter. It reports floc formation times of 12 to 13 seconds at the selected coagulant dose.

A complete report would specify the mixing regime as ASTM D2035 requires [1]: rapid mix speed and duration to disperse the reagent, slow mix speed and duration to grow the floc, and settling period before sampling, each with a stated value. It would specify the temperature at which the experiments were run, the grade and pore size of the filter medium, and the flow rate through the sorbent bed. It would define the criterion by which a floc formation time counts as acceptable and relate that criterion to residual dye.

4.5 C5: Replication and uncertainty

The manuscript reports one value per condition across all eight tables.

A complete report would present each condition in at least triplicate, with mean and standard deviation, and would apply and report a test appropriate to the sample size wherever two conditions are compared.

Replication determines which comparisons the study can make. The manuscript compares iron(III) against aluminium at mean removals of 67 and 66 per cent. Separating two conditions that differ by one percentage point requires knowledge of the measurement spread, which single measurements do not provide. Replication also identifies individual readings that diverge from their companions, which single measurements cannot distinguish from real variation.

4.6 C6: Control of experimental variables

The manuscript includes a section headed «The effect of pH on the coagulation process», which reports solution pH for each model water before and after treatment. Inlet pH ranged from 3.2 to 5.0 and outlet pH from 5.8 to 6.2 after coagulation, and from 5.3 to 6.1 after sorption.

A complete report would include a jar test series across a controlled pH range at fixed coagulant dose, run separately for each metal, with pH re-measured after coagulant addition.

Coagulation with hydrolysing metal salts is pH-dependent, because the speciation of the hydrolysis products governs which removal mechanism operates. At low pH, dissolved cationic hydrolysis species neutralise the charge on the dye. Near the solubility minimum of the metal hydroxide, precipitation removes dye by enmeshment in the growing floc. Iron and aluminium reach these conditions at different pH values, and locating each optimum fixes the pH at which performance is reported.

The pH data in the manuscript contain a pattern the report does not address. Sorption was applied to waters with inlet pH spanning 3.2 to 5.0, and outlet pH fell within 5.3 to 6.1, close to the intrinsic pH of the sorbent at 5.41. Convergence of a wide inlet range onto a narrow outlet band near the pH of the solid indicates buffering by the sorbent, which is testable.

4.7 C7: Adsorption characterisation

The manuscript reports the sorbent mass required to reach a target optical response for each model water, and an eleven-point series of optical response against sorbent mass.

A complete report would include a contact time series at fixed sorbent dose and fixed initial concentration, sampled from the first minutes to equilibrium and fitted to pseudo-first-order and pseudo-second-order kinetic models [2], with intraparticle diffusion analysis to identify whether transport into the pores or the surface interaction controls the rate [5]. It would include equilibrium isotherms across a range of initial concentrations, fitted to the Langmuir [3] and Freundlich [4] forms, with the fitted constants and goodness of fit reported.

These procedures yield two quantities the adsorption literature uses. Equilibrium contact time sets the residence time of any contactor built on the material. Maximum capacity in milligrams of dye per gram of sorbent is a property of the material, comparable against published values for other sorbents and usable in a design calculation. The manuscript reports neither.

4.8 C8: Sorbent characterisation

The manuscript reports moisture content 2.1 per cent, ash content 30 per cent, and intrinsic pH 5.41 for the sulfonated coal.

A complete report would state whether each figure was measured or taken from a supplier specification, and by what method. It would add particle size distribution, which sets external surface area and diffusion path length and therefore the kinetics, and specific surface area by nitrogen adsorption where instrument access permits. It would report the point of zero charge,

which identifies the pH above which the surface carries a net negative charge. It would report surface functional groups by infrared spectroscopy. It would include a control run of sorbent in clean water measured against the same calibration, establishing whether the material contributes to the analytical signal, which the reported ash content of 30 per cent makes relevant.

4.9 C9: Mechanistic attribution

The manuscript describes dye removal during coagulation as sorption onto hydroxide flocs and co-precipitation of aggregated dye particles. It describes sulfonated coal as a product of coal sulfonation with oleum, used industrially as a cation exchanger, and states that it possesses significant sorption capacity.

A complete report would establish which interaction retains the dye, and would test the alternatives against each other.

The question is substantive for this system. Sulfonated coal presents fixed sulfonate groups, which carry negative charge. Three of the four model waters are built from anionic dyes. A negatively charged surface and an anionic dye do not attract each other electrostatically, so retention on this material proceeds through the carbon matrix, by hydrophobic interaction and pi-stacking with the aromatic dye, by pore filling, or by bridging through multivalent cations in the ash fraction. These are separable. The point of zero charge establishes the sign of the surface charge across the working range. Uptake measured at several pH values shows whether it rises as the surface becomes less negative. Uptake measured at several ionic strengths separates electrostatic from non-electrostatic binding, since added electrolyte screens charge interactions and leaves hydrophobic interactions largely intact.

The same applies to coagulation. Charge neutralisation, sweep flocculation, and complexation between dye and metal centre have different consequences for dose, pH optimum, and sludge volume. The pH series specified under C6 separates them, as does the shape of the dose-response curve, since charge neutralisation produces restabilisation on overdose and sweep flocculation does not.

4.10 C10: Endpoint coverage and mass balance

The manuscript reports colour for every condition and pH before and after treatment.

A complete report would add chemical oxygen demand on treated and untreated samples, or total organic carbon where instrument access permits, which separates decolourisation from removal of organic load. It would add residual iron or aluminium in the treated water by atomic absorption or a colorimetric method. It would add sludge volume and dry mass from the coagulation stage.

Coagulation transfers dye from the water into a sludge and sorption transfers it onto a spent solid. Breaking the chromophore changes the appearance of the water while the organic carbon stays in solution, so colour and organic load need separate measurement. Residual coagulant metal is regulated in discharge and is inexpensive to measure. The volume and dry

mass of the sludge determine the dewatering and disposal requirement, which forms part of the operating cost.

4.11 C11: Matrix representativeness

The manuscript states that prepared solutions of commercial dyes were used, and recommends the combined process for the treatment of dye-bearing industrial effluent.

A complete report would state the difference between the tested matrix and the effluent the recommendation addresses, and would test at least one condition approaching the latter.

Reactive dye bath effluent is saline and alkaline, since dyeing uses considerable quantities of inorganic salt to promote exhaustion of the dye onto the fibre and alkali to achieve fixation, and almost all of the salt, the alkali and the unfixed dye passes into the effluent [8]. It is warm, and it carries surfactants and levelling agents together with a substantial fraction of hydrolysed dye. Each of these conditions affects coagulation and adsorption, and most reduce removal. Where real effluent is unavailable, a spiked matrix carrying sodium chloride or sodium sulfate at a realistic concentration, with one series run at alkaline pH, establishes whether performance holds.

4.12 C12: Techno-economic basis

The manuscript names low cost and availability among the design criteria for the sorbent, and states in its conclusion that sulfonated coal is inexpensive, available, and safe.

A complete report would present the cost calculation. Sorbent cost per cubic metre treated is the dose in kilograms per cubic metre multiplied by material price per kilogram, plus handling and disposal of the spent solid, plus coagulant and sludge handling from the first stage. At the dose recorded in the manuscript, the material requirement is approximately 80 kg per cubic metre of water treated.

A complete report would also address regeneration, which the manuscript states as a property of the material. Capacity retention across repeated loading and stripping cycles, thermal or chemical, determines whether the material is used once or reused.

4.13 C13: Reporting and source practice

The manuscript cites twenty-five sources, drawn from Russian-language monographs and handbooks on adsorption, colloid chemistry, ion exchange, and water treatment engineering, several published between 1973 and 1996.

A complete report would cite each listed source at a specific point for a specific claim, would include peer-reviewed journal literature alongside monographs, would include English-language sources, and would give persistent identifiers. It would retain raw measurement records and deposit the processed dataset under a permanent identifier.

The source base available through a school library and a regional university department covered the theory the study depends on. It did not cover the state of the field. Dye removal by low-cost sorbents was an active research area organised around the kinetic and isotherm

framework cited in C7, which had been established for decades, and the manuscript does not use that framework. The current literature records whether the question has already been answered, which analytical conventions make results comparable, and what published values exist for comparison.

4.14 Summary of reporting completeness

Table 1. Reporting completeness of the 2019 manuscript against the thirteen appraisal criteria.

Criterion	Reported	Additional elements in a complete report
C1 Design architecture	Three treatments, four model waters, untreated baseline, two dose-response series	Attribution of the sorbent series to a named model water; comparison across dye classes
C2 Analytical definition	Optical density, derived transmittance, derived removal percentage, clean water as comparison	Wavelength, instrument, path length, reference solution, calibration curve, linear range, detection and quantification limits, results in mg/L
C3 Dosing specification	Coagulant volume per sample volume; sorbent mass per sample volume; metal identified as Fe(III) and Al	Compound, supplier, grade, stock concentration, dose as mg metal per litre, sorbent dose as g/L, dye as mg/L
C4 Procedural specification	Sample volumes, operation sequence, filtration of floc, packed bed on glass filter, floc formation time	Rapid and slow mix speeds and durations, settling period, temperature, filter grade, bed flow rate, acceptance criterion for floc time
C5 Replication	One value per condition	Triplicate minimum, mean and standard deviation, test statistic for each comparison
C6 Control of variables	pH before and after each treatment	Jar test series across controlled pH at fixed dose, separately per metal, pH re-measured after dosing
C7 Adsorption characterisation	Sorbent mass to reach target; eleven-point mass series	Contact time series with kinetic fits, equilibrium isotherms with Langmuir and Freundlich fits, capacity in mg/g

Criterion	Reported	Additional elements in a complete report
C8 Sorbent characterisation	Moisture 2.1%, ash 30%, intrinsic pH 5.41	Provenance and method for each figure, particle size, specific surface area, point of zero charge, surface functional groups, leaching control
C9 Mechanistic attribution	Removal described as sorption on hydroxide flocs and co-precipitation; sorbent described as a cation exchanger	pH and ionic strength series to separate electrostatic from non-electrostatic retention; overdose test for restabilisation
C10 Endpoint coverage	Colour, pH	Chemical oxygen demand or total organic carbon, residual Fe and Al, sludge volume and dry mass
C11 Matrix representativeness	Prepared dye solutions	Statement of difference from real effluent; one series in a saline and alkaline matrix
C12 Techno-economic basis	Low cost and availability stated as design criteria	Cost per cubic metre from measured dose and material price; regeneration series with capacity retention
C13 Reporting and sources	Twenty-five monographic sources in Russian	Each source cited at point of use; peer-reviewed and English-language literature; persistent identifiers; retained raw records; archived dataset

5. Scope of inference supported by the 2019 design

The 2019 design supports statements about the ordering of treatments and does not support statements about the magnitude of removal.

An ordering statement compares treatments under common conditions. Systematic effects that shift every reading in the same direction cancel from a comparison, so a design with baselines, a consistent measurement procedure, and a single operator across all conditions supports ordering. The central result of the 2019 study is an ordering result, and the design supports it. Coagulation followed by sorption removed more colour than either stage applied alone, and this held across four chemically distinct dye classes.

A magnitude statement assigns a value to the removal achieved. It requires a calibrated chain linking instrument response to concentration, a stated uncertainty on that chain, and replication sufficient to place an interval around the result. A design built from single

measurements against an uncalibrated response supports the direction of an effect. Figures produced this way indicate the scale of the effect and are not performance specifications.

The study characterises dye behaviour in a clean matrix and does not establish behaviour in real effluent. It establishes decolourisation and does not establish removal of organic load. It treats the cost of the sorbent as a premise and does not establish it as a result.

Limits of this kind are normal in a first project. Each identifies an addition specified in Section 6.

6. Recommendations: specification for a redesigned study

The redesign retains the research question and the two-stage treatment architecture of the 2019 study, and changes the measurement and reporting. Recommendations are ordered by their effect on what the study can conclude.

6.1 Define the measurand before preparing solutions

Run a full spectral scan of the selected dye and record the wavelength of maximum absorbance. Prepare five to seven standards of known concentration spanning the intended working range and construct a calibration curve. Report the regression equation, the coefficient of determination, the linear range, and the limits of detection and quantification. Record the instrument, the path length, and the reference solution. Convert every subsequent measurement to concentration in mg/L.

Percentage removal then becomes a derived quantity with a defined basis, traceable through the calibration to prepared standards and reconstructable by a reader. The procedure occupies approximately one working day.

6.2 Reduce the scope to one dye

Select one dye of known structure and known molar extinction coefficient. A hydrolysed reactive azo dye is defensible for this system. Most cotton is dyed with reactive dyes, fixation efficiency is typically 50 to 80 per cent, and between a fifth and a half of applied dye is discharged unfixed or hydrolysed [8]. The same low fixation and the high water solubility of the class make it resistant to conventional coagulation, and the class is extensively represented in the published literature, which allows results to be compared against existing work.

A mixture of dyes with overlapping absorption spectra cannot be resolved into individual concentrations from a single-wavelength measurement. If the comparison across application classes is retained, each class requires its own calibrated analysis and the comparison is made in concentration units.

6.3 Express all quantities in concentration units

Report coagulant dose as milligrams of iron or aluminium per litre. Name the compound, the supplier, the grade, and the stock concentration. Report sorbent dose in grams per litre and dye content in milligrams per litre.

6.4 Replicate every condition

Run each point in at least triplicate. Report mean and standard deviation. Where two conditions are compared, apply a test appropriate to the sample size and report the statistic alongside the means. Triplicate measurement of the full condition set occupies approximately three additional laboratory sessions.

6.5 Determine the pH optimum by jar test

Run a jar test series to ASTM D2035 [1] across pH 3 to 10 at fixed coagulant dose, separately for iron(III) and for aluminium. Adjust pH with dilute acid or base and re-measure after the coagulant is added. Fix and report the mixing regime: rapid mix speed and duration, slow mix speed and duration, and settling period. Run each pH in triplicate.

The output is a curve of residual dye concentration against pH for each metal. The minimum of each curve gives the working pH for that coagulant.

6.6 Characterise the adsorption stage

Run a contact time series at fixed sorbent dose and fixed initial concentration, sampling from the first minutes to equilibrium. Fit the pseudo-first-order and pseudo-second-order forms [2] and apply intraparticle diffusion analysis [5]. Run equilibrium isotherms across a range of initial concentrations against fixed sorbent mass, and fit the Langmuir [3] and Freundlich [4] forms, reporting the fitted constants with goodness of fit.

Report equilibrium contact time and maximum capacity in milligrams of dye per gram of sorbent. The kinetic and isotherm work together occupies approximately two weeks of laboratory time.

6.7 Characterise the sorbent

Report origin, grade, and particle size distribution. Determine specific surface area by nitrogen adsorption where instrument access permits. Determine the point of zero charge. Identify surface functional groups by infrared spectroscopy. State for each reported property whether it was measured or taken from a supplier specification, and by what method. Run a control of sorbent in clean water against the same calibration.

6.8 Resolve the binding mechanism

Combine the point of zero charge from 6.7 with uptake measured across a pH series and across an ionic strength series. Rising uptake as the surface becomes less negative indicates an electrostatic component. Uptake insensitive to added electrolyte indicates hydrophobic or pore-filling retention. For the coagulation stage, use the pH series from 6.5 and the shape of the dose-response curve, testing for restabilisation on overdose.

6.9 Broaden the endpoint set

Measure chemical oxygen demand on treated and untreated samples, or total organic carbon where available. Measure residual iron or aluminium in the treated water. Measure sludge volume and dry mass from the coagulation stage.

6.10 Test a representative matrix

Repeat the combined process in a matrix carrying sodium chloride or sodium sulfate at a concentration typical of reactive dyeing, and run one series at the alkaline pH of a fixation bath. Report the conditions tested and the conditions not tested.

6.11 Establish the economic basis

Calculate sorbent cost per cubic metre treated from the measured dose and the material price, adding handling and disposal of the spent solid and the coagulant and sludge handling from the first stage. Run a regeneration series, thermal or chemical, measuring capacity retention across repeated loading and stripping cycles.

6.12 Build the source base and the data record

Work from peer-reviewed journal literature in English and Russian, located through Google Scholar and open-access repositories, with paywalled material requested through a university supervisor. Cite each source at the point of use and check it against the claim it supports at the time of writing. Maintain a bound laboratory notebook or a versioned digital equivalent containing every instrument reading, dilution, calibration, and discarded run, dated. Deposit the processed dataset and the analysis code in an open repository under a permanent identifier and an explicit licence.

6.13 Scope the problem statement to the available evidence

The 2019 introduction framed the study through contamination of the Irtysh and Ulba rivers in East Kazakhstan.

The published record for those rivers documents contamination dominated by non-ferrous metallurgy and mining in the Rudny Altai ore district. Measurements in the Ulba basin report cadmium exceeding the Kazakhstani maximum permissible concentration in 65 per cent of samples from the Ulba itself, with manganese, aluminium, iron, and cadmium exceeding WHO and US standards by factors of one to seven [6]. A ten-year retrospective across the Irtysh basin reports persistent exceedance for zinc, cadmium, and copper, with chemical oxygen demand attributed to domestic wastewater facilities [7]. Neither source identifies textile dye discharge as a contributor.

A problem statement scoped to this evidence would identify dye-bearing effluent as a problem concentrated in textile-manufacturing regions, and would identify the regional water quality problem as metal contamination from mining and metallurgy. The methods examined in the 2019 study address the former.

6.14 Redesigned measurement set

Table 2. Measurement set, as executed in 2019 and as specified in the redesign.

Quantity	2019	Redesign
Dye in solution	Optical density at unstated wavelength	Concentration in mg/L via calibration curve at stated wavelength
Coagulant dose	1.75 mL of unspecified stock into 300 mL	mg Fe or Al per litre, from named compound and stated stock concentration
Sorbent dose	g per 100 mL sample	g/L, with capacity reported in mg dye per g sorbent
pH	Recorded before and after treatment	Controlled variable across a jar test series; sorbent point of zero charge determined
Uncertainty	Not reported	Mean and standard deviation from triplicate, with test statistic per comparison
Organic load	Not measured	Chemical oxygen demand or total organic carbon, treated and untreated
Residual reagent	Not measured	Residual Fe and Al in treated water
Secondary waste	Not measured	Sludge volume and dry mass; spent sorbent mass
Matrix	Prepared dye solution	Spiked saline and alkaline matrix alongside the clean matrix
Cost	Stated as a design criterion	Calculated per cubic metre treated, including disposal and regeneration

7. Proposed work programme

The recommendations in Section 6 assemble into seven phases, ordered so that each produces the information the next requires.

Table 3. Proposed work programme.

Phase	Work	Output
1. Analytical setup	Spectral scan; calibration across five to seven standards; detection and quantification limits; precision check on replicate readings	Measurement chain traceable to prepared standards, results in mg/L
2. Sorbent characterisation	Particle size, specific surface area, point of zero charge, surface functional groups, leaching control in clean water	Material properties governing behaviour and identifying the binding mechanism
3. Coagulation optimisation	Jar test across pH 3 to 10 at fixed dose, then dose-response at the identified optimum, Fe(III) and Al separately, in triplicate	Working pH and optimum dose in mg/L for each metal; evidence separating removal mechanisms
4. Adsorption characterisation	Contact time series with kinetic fits; equilibrium isotherms with Langmuir and Freundlich fits; ionic strength and pH series	Equilibrium contact time; capacity in mg/g; the interaction responsible for uptake
5. Combined process	Coagulation at optimum followed by sorption at optimum, in triplicate, with colour, chemical oxygen demand, residual metal and sludge mass measured	Performance of the treatment train across the full endpoint set, with uncertainty
6. Matrix and durability	Combined process repeated in a spiked saline and alkaline matrix; regeneration series measuring capacity retention across cycles	Evidence on transferability to real effluent and on reuse of the sorbent
7. Assessment and release	Cost per cubic metre from measured doses; deposit of dataset and analysis code under a permanent identifier	Techno-economic figure and an open, citable research record

Total laboratory requirement is approximately ten to twelve weeks of part-time work, comparable to the 2019 study. The 2019 study allocated its effort to breadth across four dye systems. The redesign allocates it to phases 1 to 4, which establish the calibrated measurement, the material properties, the coagulation optimum and the adsorption optimum that phases 5 to 7 use.

Three procedures require equipment beyond a school laboratory: nitrogen adsorption for specific surface area, atomic absorption for residual metals, and a thermostatted shaker for isotherm work. All three are standard in a university chemistry department, which is the access the 2019 supervision arrangement provided.

8. Transferable reporting practices

The recommendations in Section 6 reduce to practices that apply to a first experimental project in any field.

1. **Fix the measurement before collecting data.** Establish how the quantity your conclusions depend on is measured, against what reference, and with what precision, before the first sample exists. A measurement basis cannot usually be retrofitted onto data already collected.
2. **Report in units another worker can use.** Concentrations, material properties, and capacities transfer between laboratories, while volumes, masses, and single-session outcomes apply only to the bench that produced them. The test is whether someone elsewhere could reproduce your conditions from the text alone.
3. **Measure each condition more than once.** Two results can be called different once the measurement spread is known, and replication is how that spread is obtained.
4. **Vary the variables you name as governing.** A factor identified as important in the introduction requires an experiment that moves it and records the response.
5. **Measure the residue as well as the target.** Removal relocates a substance into a sludge, a spent solid, or a second stream, and that stream has its own disposal requirement.
6. **Choose depth on one system over breadth across several.** A narrow question answered completely can be compared against published work and built on; a broad question answered partially usually cannot be compared against anything.
7. **Quantify the adjectives in your aim.** Words such as cheap, fast, and efficient in an objective are quantities the study has undertaken to establish.
8. Cite each source at the point of use, and check it against the claim it supports at the time of writing.
9. **Keep the raw record.** Every reading, dated and retrievable, with the reason for any discarded run.
10. **Scope each conclusion to the conditions that produced it.** The phrase under the conditions tested prevents a claim being applied to systems that were never examined.

9. Limitations of this review

This review works from the archived manuscript alone. Laboratory notebooks, raw instrument records, and calibration records from 2018 and 2019 were not available, so entries in Section 4 identify elements absent from the published record and not necessarily absent from the work as performed.

The review is conducted by the author of the work reviewed, without independent appraisal. The criteria in Section 3 were selected by the same person, and a reviewer without that relationship to the work might weight them differently.

The comparator throughout is the reporting standard of the peer-reviewed water treatment literature. Assessed against the standards applied to school research projects, the 2019 study met them, and that assessment is outside the scope of this review.

The redesign in Section 6 has not been executed. Its resource estimates are derived from the procedures specified and not from performance.

10. Conclusions

The 2019 study was built on a treatment architecture that separates the contribution of two process stages and compares them against a common baseline across four dye application classes. That architecture supports the comparative result the study reports, which is that coagulation followed by sorption removes more colour than either stage applied alone.

Statements about the magnitude of removal require elements that the published record does not contain: a defined analytical measurement, doses in concentration units, replication with a reported spread, and adsorption capacity expressed as a material property. Each of the thirteen criteria applied here identifies such elements.

The redesign specified in Section 6 retains the research question and the treatment architecture, and adds the calibrated measurement chain, the controlled pH series, the kinetic and isotherm characterisation, the broadened endpoint set, the representative matrix, and the cost calculation. It is set out as a seven-phase programme requiring approximately ten to twelve weeks of part-time laboratory work, with three procedures requiring university instrument access.

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Author note

The practices specified in Section 8 are the ones I work by. I studied in Germany and am now completing my degree in the US, and I have taken part in academic programmes at Stanford University and Johns Hopkins University, each on a full scholarship.

I am building a research programme in open biomedical data, comprising three projects, listed in the order I am developing them. Working without institutional affiliation or laboratory access, all three are computational, a constraint that shapes their design in the way that access to a university chemistry department shaped the 2019 study.

1. **A versioned, openly licensed resource linking genetic variants to drug-resistance outcomes for four newer and repurposed tuberculosis drugs: bedaquiline, clofazimine, delamanid, and linezolid.** It is built on the openly licensed CRyPTIC dataset. The first release covers bedaquiline and clofazimine, which share a resistance mechanism that current reference catalogues describe without modelling quantitatively.
2. **An open Kazakh-language biomedical language resource covering named entity recognition and terminology mapping, which does not currently exist for Kazakh.** It draws on Kazakh ICD-11 and ATC codes, English MeSH, and CC0-licensed Wikidata, and distributes pointers into source text rather than the text itself, avoiding an unsettled question in how Kazakhstani copyright law treats clinical protocols.
3. **An open, computable phytochemical and ethnopharmacological knowledge base for Central Asian medicinal plants, a region that current global databases barely cover.** It is bounded to Kazakhstani species with published chemical characterisation, and draws on Russian and German language sources not accessible to most researchers working in this area.

Each is built under the practices specified in Section 8: a defined measurand fixed before collection, versioned repositories holding raw records alongside the analysis, explicit open licensing, and datasets archived under permanent identifiers.